

BRAC UNIVERSITY
Department of Computer Science and Engineering

Examination: Quiz - 1
 Duration: 30 minutes

Semester: Spring 2025
 Full Marks: 15

CSE 340: Computer Architecture

Name: <u>Solution</u>	ID:	Section:
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1. Given that Computer has a MIPS of 5, and it requires 5 seconds to execute a specific program, CPI is 2. **Determine** the number of total clock cycles for that program. [2]

Answer: $MIPS = 5$; 5M instructions run in 1s.

$$1s - 5M \text{ ins}$$

$$\therefore 5s - (5 \times 5) \text{ s}$$

$$= 25M \text{ ins}$$

$$\text{Clock Cycle} = \text{Instruction Count} \times \text{CPI}$$

$$= 25M \times 2$$

$$= 50M = 50 \times 10^6 \text{ (Ans)}$$

2.

	Name of the Task	Number of Clock Cycle	Clock Rate (GHz)	Reference Time (s)	CPU time	SpecRatio
1	A	10×10^9	10	2	1	2
2	B	25×10^9	12	1.5	2.083	0.72
3	C	29×10^9	21	1	1.381	0.72
4	D	31×10^9	24	2	1.292	1.55

- a) Find the spec ratio for each of the Tasks. [6]

Answer:

$$\text{CPU time} = \frac{\text{Clock Cycle}}{\text{Rate (Hz)}}$$

$$\text{SPEC Ratio} = \frac{\text{Reference time}}{\text{CPU time}}$$

- b) Find the Geometric Mean. [2]

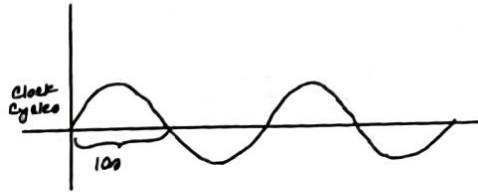
Answer:

$$G.M = \sqrt[4]{2 \times 0.72 \times 0.72 \times 1.55}$$

$$= 1.126 \text{ (Ans)}$$

3.

ADD X_{20}, X_{21}, X_{22} — 1 ADD X_{20}, X_{21}, X_{21} — 2 SUB X_{20}, X_{21}, X_{22} — 3 SUB X_{20}, X_{21}, X_{22} — 4 ADD X_{20}, X_{21}, X_{22} — 5 ADD X_{20}, X_{21}, X_{22} — 6 LD $X_{20}, 10(X_{21})$ — 7 ADD X_{20}, X_{21}, X_{22} — 8 SD $X_{21}, 10(X_{20})$ — 9	
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- Find the number of instructions in the above code. [1]
- Find the average CPI. [1]
- Find the clock cycle time. [1]
- Find the time to execute this program. [2]

- a) 9

$$b) \text{ Avg. CPI} = \frac{5 \times 3 + 2 \times 2 + 1 \times 4 + 1 \times 5}{9}$$

$$= 3.11 \text{ (Ans)}$$

IQ	CPI
Add = 5	3
Sub = 2	2
LD = 1	4
SD = 1	5

c) Clock cycle time = time taken to complete 1 clock cycle

$$= 10 + 10 = 20s \text{ (Ans)}$$

d) Execution time = Instruction count $\times$ Avg. CPI $\times$ clock cycle time

$$= 9 \times 3.11 \times 20$$

$$= 559.8 \text{ s (Ans)}$$